VARUN BANDARU
Aspiring Cloud & Security Engineer
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Profile

Security-focused Backend Engineer with hands-on experience building Zero Trust access control systems using Python, JWT (RS256), RBAC, and Docker. Strong understanding of authentication flows, policy enforcement (PDP/PEP separation), device validation, and fail-closed security design. Actively developing cloud security expertise in AWS and containerized environments.

Education

Hindu College of Engineering and Technology | Jawaharlal Nehru Technological University, Kakinada | India
Computer Science & Engineering
Expected Graduation: May2026
Cumulative GPA: 6.57 / 10
CK Jr. College | Guntur | India

Projects

Zero Trust Cloud Access Broker (ZT-CAB)

Python | FastAPI | JWT (RS256) | Docker | Microservices | RBAC | Zero Trust Architecture

- Architected and implemented a microservices-based Zero Trust Cloud Access Broker enforcing identity, device trust, and policy-based authorization before granting access to protected services.
- Built an Identity Provider issuing RSA-signed JWT tokens (RS256) with embedded role and device context.
- Designed a Device Trust Engine validating device posture before access approval.
- Implemented a Policy Decision Point (PDP) enforcing Role-Based Access Control (RBAC) across protected resources.
- Developed an API Gateway acting as Policy Enforcement Point (PEP), verifying JWT signatures, validating device trust, evaluating policy decisions, and enforcing fail-closed access control.
- Integrated centralized Audit Logging to record all access attempts (allowed/denied) for forensic traceability.
- Enforced Zero Trust principles: never trust network location, explicit verification per request, least-privilege authorization.
- Containerized all services using Docker and isolated communication within a secure internal network.
- Simulated real-world attack scenarios including unauthorized roles, invalid devices, and policy denial flows.

Experience

Security Engineering Practice (Project-Based)

- Designed and implemented identity-based access control systems aligned with Zero Trust principles.
- Developed and tested JWT-based authentication and RSA signature verification workflows.
- Implemented Role-Based Access Control (RBAC) and least-privilege authorization logic for protected resources.
- Simulated unauthorized access scenarios to validate enforcement logic and fail-closed behavior.
- Built and tested policy enforcement mechanisms using separated Policy Decision Point (PDP) and Policy Enforcement Point (PEP) architecture.
- Deployed containerized microservices using Docker with internal network isolation.
- Performed backend security validation testing including token expiry, role escalation attempts, and device trust failures.
- Worked extensively in Linux environments focusing on user permissions, access control, and process isolation.

Certifications

- FCA - FortiGate 7.6 Operator (Fortinet Training Institute)
- FCF – Introduction to the Threat landscape 3.0 (Fortinet Training Institute)
- Linux Unhatched (Cisco Networking Academy / NDG)
- Agile Scrum in Practice (Infosys Springboard)
- GEN AI & Prompt Engineering Internship (APSCHE / CSC India)

Technical Skills

Programming & Backend:
Python, FastAPI, REST APIs, JWT (RS256), Authentication & Authorization Logic, Bash
Security Engineering:
Zero Trust Architecture, RBAC, Least Privilege, Policy Enforcement (PDP/PEP), Device Trust Validation, IAM Concepts
Cloud & DevOps:
AWS (EC2 fundamentals, Security Groups), Docker, Linux (User Management, Permissions, Networking Basics)
Networking & Infrastructure:
TCP/IP Fundamentals, Firewalls, FortiGate Basics
Tools:
Git, GitHub, MySQL, VS Code